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 SURVEY

How Deep Learning Sees the World: A Survey on Adversarial Attacks & Defenses

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ABSTRACT Deep Learning is currently used to perform multiple tasks, such as object recognition, face recognition, and natural language processing. However, Deep Neural Networks (DNNs) are vulnerable to perturbations that alter the network prediction, named adversarial examples, which raise concerns regarding the usage of DNNs in critical areas, such as Self-driving Vehicles, Malware Detection, and Healthcare. This paper compiles the most recent adversarial attacks in Object Recognition, grouped by the attacker capacity and knowledge, and modern defenses clustered by protection strategies, providing background details to understand the topic of adversarial attacks and defenses. The new advances regarding Vision Transformers are also presented, which have not been previously done in the literature, showing the resemblance and dissimilarity between this architecture and Convolutional Neural Networks. Furthermore, the most used datasets and metrics in adversarial settings are summarized, along with datasets requiring further evaluation, which is another contribution. This survey compares the state-of-the-art results under different attacks for multiple architectures and compiles all the adversarial attacks and defenses with available code, comprising significant contributions to the literature. Finally, practical applications are discussed, and open issues are identified, being a reference for future works.

INDEX TERMS Adversarial attacks, adversarial defenses, datasets, evaluation metrics, review, vision transformers.

I. INTRODUCTION

Machine Learning (ML) algorithms have been able to solve various types of problems, with Deep Neural Networks (DNNs) [1] achieving results similar to, or better than, humans in multiple tasks, such as Object Recognition [2], [3], Face Recognition [4], [5], and Natural Language Processing [6], [7]. These networks have also been employed in critical areas, such as Autonomous Driving [8], [9], Malware Detection [10], [11], and Healthcare [12], [13], whose application and impaired functioning can severely impact their users.

DNNs results suggested that these networks could correctly generalize in the local neighborhood of an input

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(image), which motivated their adoption and integration in real-time image analysis, making malicious entities target these techniques. Recently, Szegedy et al. [14] found that small perturbations in images can entirely alter DNNs prediction, without changing Human prediction and significantly increasing the confidence of erroneous output. Another early study [15] distinguishes four types of adversaries depending on the amount of information they can access, which was relevant to cluster the different types of attacks, and suggested that the strength of the attack is diminished in real scenarios, requiring queries that are usually limited or very expensive [16]. The perturbations effect has been analyzed with more emphasis on Object Recognition, being one of the inception areas for Deep Learning.

In this survey, we present the most recent adversarial attacks and defenses in the Object Recognition area. The

defense mechanisms [17], [18], [19], [20], which we clustered on six different domains, can reliably defend against legacy attacks [14], [21]. However, there is no assurance about the effective robustness of DNNs and if they can be trusted in critical areas, clearly raising the need to make the DNNs inherently robust or easy to be updated every time a new vulnerability is encountered.

Previous surveys [22], [23], [24] do not provide a state-of-the-art comparison, area-specific metrics, and architectures commonly used, hindering the initiation among new researchers. This survey provides a comprehensive review of the legacy and recent advances, such as Adversarial Purification and Adversarial Attacks in Vision Transformers (ViTs), presents datasets and metrics in this context, provides results for multiple datasets, using the *de facto* standard attack (Auto-Attack [25]), and summarizes approaches that have official or unofficial implementations (duly identified) to stimulate the research of new mechanisms. This survey intends to answer and was guided by the following research questions:

- **RQ1:** Which are the most recent developments regarding Vision Transformers?
- **RQ2:** Which metrics are used to evaluate the effect of the attacks and robustness of models/defenses?
- **RQ3:** What are the state-of-the-art results and architectures in Adversarial Attacks and Defenses?
- **RQ4:** Which practical applications and possible future directions in the topic of Adversarial Attacks and Defenses in Object Detection can we follow?

The papers considered in this study were all included in proceedings of international CORE A* and A conferences or published in Q1 and Q2 journals and the initial collection was built by conducting searches with the keywords “*Adversarial*”, “*Attack*”, “*Defense*”, “*Adversarial Attack*”, and “*Robustness*”, starting from 2013 (inception of this topic). The search engines were IEEE Xplore, Google Scholar, ACM Digital Library, and Elsevier. The inclusion criteria admitted papers that cited the key works [14], [15], full papers published in journals and conferences, written in English, and articles that address an adversarial attack/defense on state-of-the-art datasets. The exclusion criteria ruled out duplicate papers, whitepapers, articles shorter than two pages, studies without full text available, no proposal of new attacks or defenses, no evaluation on the mentioned datasets, and containing only non-reproducible experiments. Papers that were not related to the topic in question were excluded by reading the abstract, introduction, and conclusion. The main contributions of this work can be summarized as follows:

- We present the most recent adversarial attacks grouped by the adversary capacity, accompanied by an illustration of the differences between black-box and white-box attacks;
- We propose six different domains for adversarial defense grouping, assisted by exemplary figures of each of

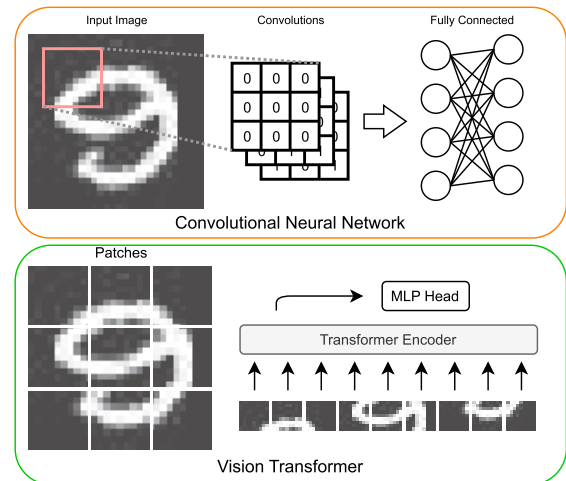


FIGURE 1. Schematic example of the convolutional neural networks (top) and vision transformers (bottom) to classify images.

these domains, and describe the effects of adversarial examples in ViTs;

- We detail the most widely used metrics and datasets, present state-of-the-art results on CIFAR-10, CIFAR-100, and ImageNet, and propose directions for future works.

The remaining of the paper is organized as follows: Section II provides background information; Section III compares this review with others; Section IV presents the set of adversarial attacks; Section V shows a collection of defenses to overcome these attacks; Section VI describes the advances in ViTs; Section VII displays the commonly used datasets; Section VIII lists and elaborates on metrics and presents state-of-the-art results; and Section IX presents future directions, with the concluding remarks included in Section X.

II. BACKGROUND FOR ADVERSARIAL ATTACKS

A. NEURAL NETWORK ARCHITECTURES

When an input image is fed into a CNN [26], it is converted into a matrix, or a set of matrices, containing the numeric values for each color channel. Then, the Convolutional layers apply filters to these matrixes and calculates a set of reduced-size features, which are fed into the Fully Connected layers that classify the provided image. ViT [27] gives a pre-processed image that is divided into Patches, which are crops of the original image, and their ordered position to the Transformer Encoder. The output resulting from this encoder is fed into the Multi-Layer Perceptron (MLP) Head that classifies the image. Figure 1 shows a schematic overview of CNNs and ViTs. For a complete explanation of the functioning of CNNs and ViTs we refer to [1] and [27], respectively.

B. ADVERSARIAL EXAMPLE

Misclassification can be justified if the object contained in the image is not visible even to Humans, yet adversarial examples

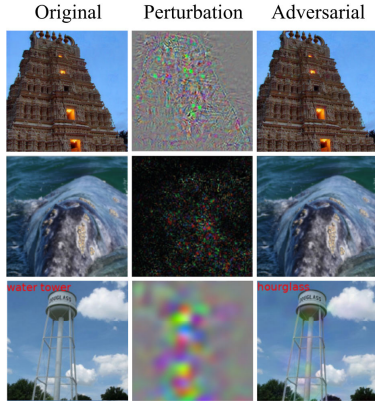


FIGURE 2. Adversarial examples created using different adversarial attacks. The images were withdrawn from [14], [28], and [29] and resized for better visualization.

do not fit this scope. These examples add a perturbation to an image that causes the DNNs to misclassify the object in the image, yet Humans can correctly classify the same object. The adversarial attacks focus on identifying the perturbations that make DNNs misclassify, with their calculation being an optimization problem formally defined as:

$$\arg \min_{\delta \mathbf{X}} \|\delta \mathbf{X}\| \quad \text{s.t.} \quad f(\mathbf{X} + \delta \mathbf{X}) = \mathbf{Y}^*, \quad (1)$$

where f is the classifier, $\delta \mathbf{X}$ is the perturbation, $\mathbf{X}$ is the original/benign image, and $\mathbf{Y}^*$ is the adversarial output. Furthermore, the adversarial example is defined as:

$$\mathbf{X}^* = \mathbf{X} + \delta \mathbf{X}, \quad (2)$$

where $\mathbf{X}^*$ is the adversarial image. Figure 2 displays adversarial examples generated using different attacks. Limited-memory Broyden-Fletcher-Goldfarb-Shanno (L-BFGS) [14], DeepFool [28], and SmoothFool [29] correspond to the first, second, and third rows. The L-BFGS applies noise to almost the entirety of the adversarial image, the DeepFool only perturbs the area of the depicted whale but not all the pixels in that area, and the SmoothFool slightly disturbs the pixels in the area of the image, displaying the evolution of adversarial attacks in decreasing order of detectability. To limit the amount of noise, the adversarial attacks are divided into L_0 , L_2 , and L_p norms, known as *Vector Norms*, and common terminologies in the context of adversarial examples are defined in Table 1.

C. VECTOR NORMS AND ϵ CONSTRAINT

Vector Norms are functions that take a vector as input and output a positive value (scalar), which are essential to ML and allow the backpropagation algorithms to compute the loss value as a scalar. These functions are known as the p -norm, and the considered values are 0, 2, and ∞ . L_0 norm consists of counting the number of non-zero elements in the vector, L_2 norm, also known as the Euclidean distance, measures the vector distance to the origin, and L_∞ norm represents the maximum hypothetical value that p can have and returns the absolute value of the element with the largest magnitude.

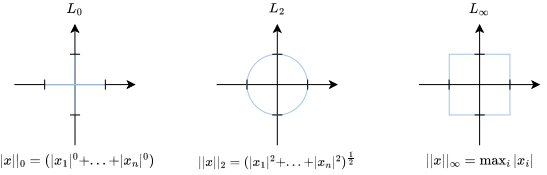


FIGURE 3. Geometric representation and formal definition of the L_0 , L_2 , and L_∞ norms, from left to right, respectively, with x_i referring to each element of the vector x .

TABLE 1. Common terminologies used in the context of adversarial attacks and their definition.

Terminology	Definition
Clean Example	Image without noise from a dataset
Adversarial Example	Image with a perturbation that fools a classifier
Perturbation	Set of changes/transformations applied to a image
Adversarial Attack	Technique used to calculate the perturbation
Transferability	Capability of an adversarial example being transferred from a known network to an unknown network

A geometric representation of the area of exploitation and a formal definition for the three considered p -norm is displayed in Figure 3.

One relevant property of the p -norm is: the higher p is, the more important the contribution of large errors; the lower p is, the higher the contribution of small errors. This translates into a large p benefiting small maximal errors (minimal perturbations along multiple pixels) and a small p encouraging larger spikes in fewer places (abrupt perturbations along minimal pixels). Therefore, L_2 and L_0 attacks have greater detectability than L_∞ attacks, with the latter being more threatening.

Another constraint normally seen is ϵ , which is a constant that limits the amount of noise that can be added to an image, and it is usually a tiny number that varies depending on the dataset, decreasing when the task increases in difficulty. According to the literature, for MNIST, $\epsilon = 0.1$, for CIFAR-10 and CIFAR-100, $\epsilon = 8/255$, and for ImageNet, $\epsilon = 4/255$.

D. ADVERSARY GOALS AND CAPACITY

Besides the restriction imposed by the different Vector Norms, the adversarial attacks are also divided by their impact on the networks, depending on the goals of the attacker, with the following designation: 1) **Confidence Reduction**, the classifier outputs the original label with less confidence; 2) **Untargeted**, the classifier outputs any class besides the original label; and 3) **Targeted**, the classifier outputs a particular class besides the original label.

As defined by Papernot et al. [15], the attackers can have access to: 1) training data and network architecture; 2) only network architecture; 3) only training data; 4) an oracle that replies to all the inputs given; and 5) only have pairs of input and corresponding output (samples). However, to simplify this classification, these capacities were divided into: 1) **White-box**, which considers that the attacker has access to both architecture and data; and 2) **Black-box**, when the attacker can only access samples from an oracle or pairs of input and output.

TABLE 2. Characteristics shown in state-of-the-art surveys on adversarial attacks.

Survey	Year	White & Black-Box	Survey Comparison	Grouping of Defenses	Future Directions	Datasets Overview	Metrics and Architectures	State-of-the-art Comparison	Vision Transformers
Akhtar and Mian [30]	2018	✓	×	✓	✓	×	×	×	×
Qiu <i>et al.</i> [31]	2019	✓	×	×	×	×	×	×	×
Serban <i>et al.</i> [32]	2020	✓	×	✓	✓	×	×	×	×
Xu <i>et al.</i> [33]	2020	✓	×	✓	×	×	×	×	×
Sadeghi <i>et al.</i> [34]	2020	✓	✓	✓	✓	✓	×	×	×
Chakraborty <i>et al.</i> [35]	2021	✓	×	×	×	×	×	×	×
Long <i>et al.</i> [22]	2022	✓	✓	×	✓	×	×	×	×
Liang <i>et al.</i> [23]	2022	✓	×	✓	✓	×	×	×	×
Zhou <i>et al.</i> [24]	2022	✓	×	✓	✓	✓	×	×	×
Mi <i>et al.</i> [36]	2023	✓	×	✓	×	×	×	×	×
Amritasree <i>et al.</i> [37]	2023	✓	×	✓	✓	×	×	×	×
This survey	2024	✓	✓	✓	✓	✓	✓	✓	✓

III. RELATED SURVEYS

Akhtar and Mian [30], initially surveyed adversarial attacks in computer vision, extensively referring to attacks for classification, providing a brief overview of attacks beyond the classification problem, presenting a set of attacks performed in the real world, providing insight into the existence of adversarial examples, and clustering defenses in three categories. From a broader perspective, Liu *et al.* [38] studies the security threats and possible defenses in ML scope, considering the different phases of an ML algorithm, and provides insight on the currently used techniques. Chakraborty *et al.* [35] provides insight into commonly used ML algorithms, presents the adversary capabilities and goals, cluster adversarial attacks based on the stage of the attack, and introduces relevant defenses used in adversarial settings.

Focusing more on the object recognition task, Serban *et al.* [32] extensively analyzed the adversarial attacks and defenses, providing conjectures for the existence of adversarial examples and evaluating the capacity of adversarial examples transferring between different DNNs. Qui *et al.* [31] explains background concepts in adversarial attacks, displays practical applications, presents the most relevant adversarial defenses, and illustrates a set of attacks divided by the stage in which they occur, referring to the most relevant attacks. Sadeghi *et al.* [34] summarizes the basic concepts, analyzes datasets and ML architectures, providing awareness of the adversary strategies and the defense responses to them, and proposes a comprehensive system-drive taxonomy for the latter. Long *et al.* [22] discusses a set of preliminary concepts of Computer Vision and adversarial context, providing a set of adversarial attacks grouped by adversary goals and capabilities, and propose a set of research directions that readers can use to continue the development of robust networks. Liang *et al.* [23] discuss the most significant attacks and defenses in the literature, with the latter being grouped by the underlying technique, finishing with the presentation of existing challenges in the adversarial context.

Xu *et al.* [33] also describes background concepts, summarizes the most relevant adversarial attacks at the time of that work, presenting physical world examples, classify defenses by the underlying methodology, and outline adversarial attacks in graphs, text, and audio networks, culminating in the possible applications of these attacks. More recently,

Zhou *et al.* [24] provides insight into Deep Learning and Threat Models, focusing on the Cybersecurity perspective. Therefore, the authors identify multiple stages based on Advanced Persistent Threats, explain which adversarial attacks and defenses are employed in each stage, present the commonly used datasets in adversarial settings, and provide a set of future directions from a Cybersecurity perspective.

Mi *et al.* [36] and Amritasree *et al.* [37] evaluate the recent adversarial attacks and defenses, providing an extensive description of the different types of adversarial attacks, that specifically focus on Object Detection. The latter [37] includes a set of future research directions.

The literature clarifies that some of the specialized concepts of the area, such as adversary goals and capabilities and adversarial attacks and defenses, have already solidified and become standard. However, the datasets, architectures, and metrics usage still lack standardization. In an attempt to do so, this survey analyzes datasets, architectures, and metrics to provide insight to novice researchers. Additionally, it presents state-of-the-art results and provides recent advancements in adversarial attacks on ViTs, two contributions of this survey. Concluding with a set of future directions that researchers and practitioners can follow to start their work, which is the main contribution. A comparison between the several surveys discussed in this section (with the work reflected herein) is summarized in Table 2.

IV. ADVERSARIAL ATTACKS

Adversarial attacks are commonly divided by the amount of knowledge the adversaries have access to, white-box and black-box, as can be seen in Figure 4. Furthermore, the development of defenses is highly motivated by the proposal of new attacks, with the same being true for the attacks, suggesting that some defenses might have overcome some attacks and vice-versa. For example, Carlini and Wagner attack overcomes the Defensive Distillation methodology.

A. WHITE-BOX SETTINGS

Adversarial examples were first proposed by Szegedy *et al.* [14], which discovered that DNNs do not generalize well in the vicinity of an input, leading to the proposal of **L-BFGS**, which raised awareness in the scientific community for this generalization problem. L-BFGS is a computationally

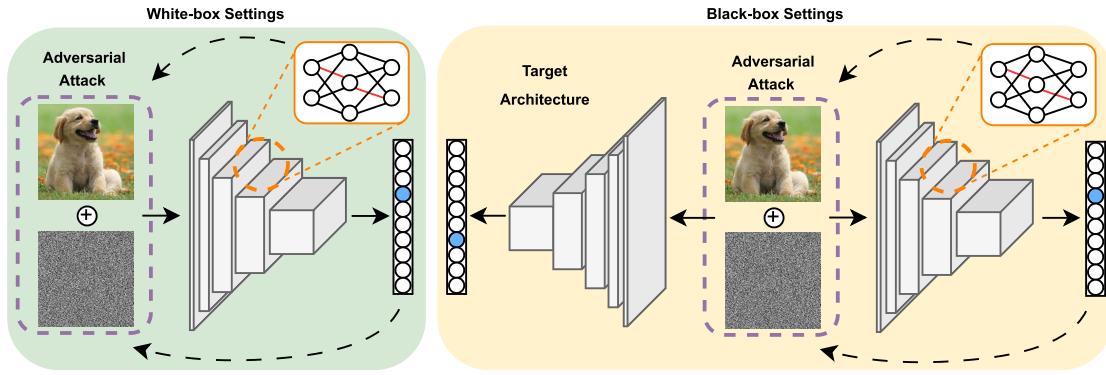


FIGURE 4. Schematic overview of an adversarial attack under white-box settings (left) and black-box settings (right). The first one uses the classifier predictions and network gradients to create perturbations (similar to noise), which can fool this classifier. These perturbations are added to the original images, creating adversarial images, which are fed to the network and cause misclassification. In the black-box settings, the same process is applied to a known classifier, and the obtained images are used to attack another classifier (represented as target architecture).

expensive method, which motivated **Fast Gradient Sign Method (FGSM)** [21], a one-step method to find adversarial examples, calculated using the model cost function, the gradient, and the radius epsilon. This attack is formally defined as:

$$\mathbf{x} - \epsilon \cdot \text{sign}(\nabla \text{loss}_{F,t}(\mathbf{x})), \quad (3)$$

where $\mathbf{x}$ is the original image, ϵ is the amount of changes to the image, and t is the target label. The value for ϵ should be very small to make the attack undetectable.

Jacobian-based Saliency Maps (JSM) [15] explore the forward derivatives to calculate the model gradients, replacing the gradient descent approaches, and discover which input regions are likely to yield adversarial examples. Then, it constructs the adversarial saliency maps, displaying the features the adversary must perturb, with only the adversarial examples correctly classified by humans being used to fool neural networks. **Gradient Aligned Adversarial Subspace (GAAS)** [39] estimates the dimensionality of the adversarial subspace using the first-order approximation of the loss function, proving to be the most successful at finding many orthogonal attack directions, which indicates that neural networks generalize linearly. Croce and Hein [40] noticed that perturbations generated by l_0 attacks are sparse and by l_∞ attacks are smooth on all pixels, proposing **Sparse and Imperceptible Adversarial Attacks (SIAA)**, which creates sporadic and imperceptible perturbations by applying the standard deviation of each color channel in both axis directions, calculated using the two immediate neighboring pixels and the original pixel.

DeepFool [28] is an iterative attack that stops when the minimal perturbation, corresponding to the vector orthogonal to the hyperplane representing the decision boundary, is found. **SmoothFool (SF)** [29] is an iterative algorithm that uses DeepFool to calculate the initial perturbation and smoothly rectifies the resulting perturbation until the adversarial example fools the classifier, providing smoother perturbations that improve the transferability of adversarial examples, whose impact varies with the different categories

in a dataset. **Projected Gradient Descent (PGD)** [41] is an iterative attack that uses saddle point formulation, an inner maximization problem to find an adversarial example that achieves high loss and an outer minimization problem to find model parameters that minimize the first, to find a strong perturbation. The saddle point problem used by PGD is defined as:

$$\min_{\theta} \rho(\theta), \text{ where } \rho(\theta) = \mathbb{E}_{(\mathbf{x}, y) \sim \mathcal{D}} \left[\max_{\delta \in \mathcal{S}} L(\theta, \mathbf{x} + \delta, y) \right], \quad (4)$$

where $\mathbf{x}$ is the original image, y is the corresponding label, and $\mathcal{S}$ is the set of allowed perturbations δ . Motivated by the inability to achieve a high transferability, the **Momentum Iterative FGSM (MI-FGSM)** [42] introduces momentum, a technique for accelerating gradient descent algorithms, into the Iterative FGSM (I-FGSM), showing that the attack transferability almost doubles.

Carlini and Wagner (C&W) [43] attack is one of the most powerful attacks, which uses three different vector norms: 1) L_2 , a smoothing of clipped gradient descent approach, displaying low distortion; 2) L_0 , an iterative algorithm that, at each iteration, fixes the pixels that do not have much effect on the classifier and finds the minimum amount of pixels that need to be altered; and 3) L_∞ , an iterative algorithm with an associated penalty, penalizing every perturbation that exceeds a predefined value, formally defined as:

$$\min \quad c \cdot f(\mathbf{x} + \delta) + \sum_i [(\delta_i - \tau)^+], \quad (5)$$

where δ is the perturbation, τ is the penalty threshold (initially 1, decreasing in each iteration), and c is a constant. The value for c starts as a very low value (e.g., 10^{-4}), and each time the attack fails, the value for c is doubled. If c exceeds a threshold (e.g., 10^{10}), it aborts the search. **Meta Gradient Adversarial Attack (MGAA)** [44] can be integrated into any existing gradient-based attack method to improve cross-model transferability, consisting of multiple iterations, and various models that are used to generate adversarial perturbations, which are added to the previously

generated perturbations. Using multiple models simulates both white- and black-box settings, making the attacks more successful.

AdvGAN [45] uses Generative Adversarial Networks (GAN) [46] to create realistic adversarial examples that have high attack success rate. The generator receives the original instance and creates a perturbation, the discriminator distinguishes the original instance from the perturbed instance, and the target neural network is used to measure the distance between the prediction and the target class. Recently there was a shift into natural Unrestricted Adversarial Examples (UAEs), breaking the l_p perturbation bounds while remaining semantically plausible, considered by **AdvDiffuser** [47], which generates UAEs using diffusion models that are more natural and stronger than the state-of-the-art. In an attempt to exploit the vulnerabilities of detector networks, **Adversarial Manipulation Generation** (AMG) [48] is an attack that uses spatial and frequency features to generate adversarial examples that fool the detector networks, having high success rate.

Another topic that is being discussed under the context of white-box is the existence of backdoor attacks, by including poisoned data in the training process. Chen et al. [49] proposes a stealthier backdoor attack using input-instance-key and pattern-key strategies. Liu et al. [50] performs a backdoor attack not by manipulating the initial training but by doing an additional train that takes only a few hours. **Bad-Nets** [51] considers the creation of DNNs from outsourcing and third-party cloud providers and Weng et al. [52] evaluates the trade-off between adversarial resilience and backdoor robustness, by conducting extensive experiments.

Summary: The white-box attacks assume that the attacker has access to information regarding the model, such as weights and data. Therefore, they are the most powerful type of attack and can, the majority of times, completely fool the models. However, these attacks are model-specific and do not transfer to different models, having to be executed again for each model.

B. UNIVERSAL ADVERSARIAL PERTURBATIONS

Moosavi-Dezfooli et al. [53] discovered image-agnostic perturbations, labeled as **Universal Adversarial Perturbations** (UAPs), that generalize well and cause misclassification with high probability, by searching for a vector of perturbations that causes misclassification in almost all the data drawn from a distribution of images. The optimization problem that Moosavi-Dezfooli et al. are trying to solve is the following:

$$\Delta v_i \leftarrow \arg \min_{\mathbf{r}} \|\mathbf{r}\|_2 \quad \text{s.t.} \quad \hat{k}(\mathbf{x}_i + \mathbf{v} + \mathbf{r}) \neq \hat{k}(\mathbf{x}_i), \quad (6)$$

where Δv_i is the minimal perturbation to fool the classifier, $\hat{k}$ is the classifier, $\mathbf{r}$ is a perturbation vector that sends $\mathbf{x}_i + \mathbf{v}$ to the decision boundary, $\mathbf{v}$ is the universal perturbation, and $\mathbf{x}_i$ is the original image. This optimization problem is calculated for each image in a dataset, and the vector containing the universal perturbation is updated.

The **Universal Adversarial Networks** (UAN) [54] are GANs that use UAPs as the training set and are capable of outputting perturbations for any given input that fool classifiers, demonstrating more outstanding results than the original UAPs. Based on UAP generation problems, gradient vanishing and local optima, **Stochastic Gradient Aggregation** (SGA) [55] uses multiple iterations of inner pre-search and aggregates all the inner gradients as a one-step gradient estimation. UAPs are highly dependent of prior data, leading to the proposal of data-free UAP methods that rely on random noises and maximize the CNN activation, **TRM-UAP** [56] is a data-free UAP generator that does not depend on any real data through truncated ratio maximization. Exploring the vulnerabilities of different downstream tasks, **AdvEncoder** [57] constructs a UAP that is able to fool all the downstream tasks, providing a new insight for how adversarial attacks can be applied.

Kurakin et al. [58] initially demonstrates that adversarial examples exist in the physical world, by generating adversarial examples and printing them on paper. **Basic Iterative Method** (BIM) applies the FGSM multiple times with a small step size between iterations and clips the intermediate values after each step, and **Iterative Least-likely Class Method** (ILCM) uses the least-likely class, according to the prediction of the model, as the target class and uses BIM to calculate the adversarial example that outputs the target class. **Adversarial Camouflage** (AdvCam) [59] crafts physical-world adversarial examples that are legitimate to human observers, using the target image, region, and style to perform a physical adaptation (creating a realistic adversarial example). Physical adversarial attacks also use printed adversarial patches/camouflages to alter the appearance of the target object, while remaining semantically plausible, **Reflected Light Attack** (RFLA) [60] generates adversarial examples by placing a color transparent plastic sheet and a paper cut aligned in front of a mirror to generate multiple colored geometries on the target object. Wei et al. [61] proposes a **Unified Adversarial Patch** that performs cross-modal, both visible and infrared, physical attacks with a single patch.

Summary: The goal of Universal Adversarial Perturbations is to create unique perturbations that have effect on multiple models and multiple images, and they are a subgroup of the black-box attacks. The same perturbation can be added to different images to perform an attack for different models, yet these perturbations are weaker than white-box but more transferable.

C. BLACK-BOX SETTINGS

Specifically considering the black-box setup, Limited Queries and Information Attack (LQIA) [62] proposes three different realistic approaches: 1) query-efficient algorithms, using Natural Evolutionary Strategies to estimate the gradients used to perform the PGD attack; 2) alternate between blending in the original image and maximizing the likelihood of the target class; and 3) include noise

robustness to mount a targeted attack. Under the same problem, **Subspace Activation Evolution Strategy** (SAES) [63] proposes an optimization method to capture the most promising direction, in a query-limited setting, generating a distribution of adversarial examples.

Feature-Guided Black-Box (FGBB) [64] views the creation of adversarial examples, using Scale Invariant Feature Transform, as a two-player game, where the first minimizes the distance and the second has multiple roles, leading to minimal adversarial examples. Another approach based on creating minimal adversarial examples using the decision of a model is the **Boundary Attack** [65], which repeatedly decreases the distance between the adversarial image and the target image. Similarly, **HopSkipJump** [66] is a decision-based attack that uses fewer queries, based on an optimized gradient estimation. The challenge of Black-box settings has led to multiple prospects, such as synthetic data [67], fluctuation phenomenon [68], and input transformation [69].

Square Attack [70] is an attack that uses a randomized search scheme that selects localized square-shaped updates in random positions, causing the perturbation to be situated at the decision boundaries. Further exploring the advantages of features, **Transferable Targeted Adversarial Attack** (TTAA) [71] captures the distribution information of the target class from label- and feature-wise perspective to generate highly transferable adversarial examples and **Style-Less perturbation** (StyLess) [72] uses stylized networks, encoding different style features, by perturbing an adaptive instance normalization. **Feature Importance-aware Attack** (FIA) [73] considers the object-aware features that dominate the model decisions and identifies the essential features, generating highly transferable adversarial examples when extracting the feature importance from multiple classification models.

Zanddizari et al. [74] sets a threshold to choose large sparse (limiting the amount of pixels that can be altered) components and perturb them either randomly or in the direction of another class. However, the limitation of sparse attacks causes the generation of evident perturbations, reducing the stealthiness of these attacks. To minimize this effect, **Multi-objective Sparse Attacks** (MSA) [75] efficiently minimizes the number of modified pixels and the size of perturbations, increasing the stealthiness and having state-of-the-art results.

Inspired on the difficulties of transferring adversarial examples from CNNs to ViTs and the success of ensemble attacks, **AdaEA** [76] adaptively controls the fusion of the outputs from each model toward the adversarial objective and **LEA**² [77] considers three models with non-overlapping vulnerable frequency regions to cover a large vulnerable subspace. **Momentum Integrated Gradients** (MIG) [78] uses the Integrated Gradients and Momentum updating strategy to generate adversarial examples with high success against ViTs and transferability across the ViTs and CNNs.

Summary: There has been a recent focus on developing stronger Black-box attacks, since these are the more realistic

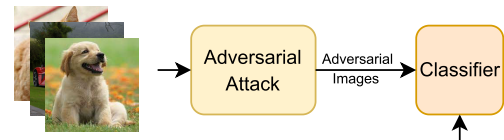


FIGURE 5. Schematic overview of adversarial training. A subset of the original images of a dataset is fed into an adversarial attack (e.g., PGD, FGSM, or C&W), which creates adversarial images. Each batch contains original and adversarial images, with the Classifier being normally trained.

ones, and assume that the attacker only has access to queries or pairs of input and respective prediction. Despite this, these attacks have lower impact on the performance of the model and can more easily be outperformed by defenses.

D. AUTO-ATTACK

Auto-Attack [25] is capable of testing adversarial robustness in a parameter-free, computationally affordable, and user-independent way, by proposing two variations of PGD that overcome sub optimal step sizes of the objective function, APGD-CE and APGD-DLR, with PGD using Cross-Entropy (CE) and Difference of Logits Ratio (DLR) loss, respectively. These new PGD variations are combined with Fast Adaptive Boundary (FAB) [79], which minimizes the norm of the adversarial perturbations, and Square Attack [70], a query-efficient black-box attack, to create Auto-Attack. Given the need for a computationally affordable approach, the FAB attack in Auto-Attack is a targeted version of FAB, which is both computationally and memory-wise plausible in datasets with an increased number of classes (e.g., CIFAR-100 and ImageNet). Auto-Attack is an ensemble of attacks with important fundamental properties: APGD is a white-box attack aiming at any adversarial example within an L_p -ball, FAB minimizes the norm of the perturbation necessary to achieve a misclassification, and Square Attack is a score-based black-box attack for norm bounded perturbations which use random search and do not exploit any gradient approximation, competitive with white-box attacks [70].

V. ADVERSARIAL DEFENSES

The countermeasures are divided by their practical point of view, namely how quickly can they be adopted to the current models, and are an adaptation of previously presented taxonomies [32], [38].

A. ADVERSARIAL TRAINING

Szegedy et al. [14] proposed that training on a mixture of adversarial and clean examples could regularize a neural network, as shown in Figure 5. Goodfellow et al. [21] evaluated the impact of **Adversarial Training** as a regularizer by including it in the objective function, showing that this approach is a reliable defense that can be applied to every neural network. Some preliminary works evaluated how different image preprocessing techniques could influence the robustness, Gu and Rigazio [80] used noise injection, autoencoder, and denoising autoencoder and Zheng et al. [81]

used compression, rescaling, and cropping, denominated as **Image Processing**, discovering that adversarial noise mainly exists in high-frequency domain.

Kurakin et al. [82] performs adversarial training in more massive datasets (ImageNet), for the first time, displaying that the robustness significantly increases when using one-step methods. Training a model using one-step attacks and ground-truth labels leads to better accuracy on the adversarial images than on the clean images, an effect denominated as **Label Leaking**, suggesting that the adversarial training should not make use of the ground-truth labels. **Ensemble Adversarial Training** (EAT) [83] uses adversarial examples crafted for other pretrained models to augment training data to increase resilience to black-box attacks. To avoid the Label Leaking effect, Zhang and Wang [19] suggest that adversarial examples are generated through **Feature Scattering** (FScatter) in the latent space, by maximizing the feature-matching distance between the clean and perturbed examples, and the DNN performs standard adversarial training. To tackle the same issue, **Unsupervised Adversarial Training** (UAT) [84] suggests the training with unlabeled data considering two different approaches: (i) UAT with Online Target (UAT-OT) that minimizes a differentiable surrogate of the smoothness loss and (ii) UAT with Fixed Targets (UAT-FT) that trains an external classifier to predict the labels on the unsupervised data and uses its predictions as labels. Considering that disturbing labels can also improve accuracy, **Helper-based Adversarial Training** (HAT) [85] incorporates additional wrongly labeled examples during training, achieving a good trade-off between accuracy and robustness.

Shared Adversarial Training [86] uses gradients to create a shared perturbation for a mini-batch of images, which are accumulated and clipped to constrain them into a predefined magnitude, creating the adversarial examples used for adversarial training. Other approach that uses custom-made adversarial examples is **Defense against Occlusion Attacks** (DOA) [87], which is a defense mechanism that includes physically realizable attacks, “normal” in the real world (*e.g.*, eyeglasses and stickers), and applies standard adversarial training.

TRadeoff-inspired Adversarial DEFense via Surrogate-loss minimization (TRADES) [88] is inspired by the presumption that robustness can be at odds with accuracy [89], [90], encouraging the natural error to be optimized and minimizing the distance between prediction of natural and adversarial examples. However, this approach still lacks differentiating misclassified from correctly classified examples, thus **Misclassification Aware adversarial Training** (MART) [91] explicitly differentiates these examples during training by introducing a misclassification aware regularization term.

Despite the popularity of adversarial training to defend models, it has a high cost of generating strong adversarial examples, **Free Adversarial Training** (Free-AT) [92] uses the gradient information when updating model parameters to generate the adversarial examples, eliminating the previ-

ously mentioned overhead. Another optimization regarding the computational costs, consists in only considering the first layer of a network for forward and backpropagation, effectively reducing the amount of propagation to one in each update, named as **You Only Propagate Once** (YOPO) [93], due to the adversary update being only related to the first layer. Exploring the different layers of a network, **Latent Adversarial Training** (LAT) [94] consists of finetuning adversarial-trained models to ensure robustness at the latent level, because the latent layer is significantly vulnerable to adversarial perturbations of small magnitude. Instead of using computational costly methods, **Cheap Adversarial Training** (Cheap-AT) [95] proposes the use of weaker and cheaper adversaries (FGSM) combined with random initialization to train robust networks effectively. Inspired by the results from Cheap-AT, **Fast Adversarial Training** (FAT) [96] performs randomized smoothing to optimize the inner maximization problem efficiently, and proposes a new initialization strategy, named backward smoothing, that helps to improve the stability and robustness of a model using single-step robust training methods.

The proposal of **Smooth Adversarial Training** (SAT) [97] considers the evolution normally seen in curriculum learning, where the difficulty increases with time (age), using two difficulty metrics based on the maximal Hessian eigenvalue (H-SAT) and the softmax Probability (P-SAT), which are used to stabilize the networks for large perturbations while having high clean accuracy. In the same context, **Friendly Adversarial Training** (Friend-AT) [98] minimizes the loss considering the least adversarial data (friendly) among the adversarial data that is confidently misclassified. **Oracle-Aligned Adversarial Training** (OA-AT) [99] creates a model that is robust to high perturbation bounds by aligning the network predictions with ones of an Oracle during adversarial training, using Learned Perceptual Image Patch Similarity [100] to generate Oracle-Invariant attacks and convex combination of clean and adversarial predictions as targets for Oracle-Sensitive samples.

Bilateral Adversarial Training (BAT) [101] proposes to perturb the images and labels (adversarial labels) during the training phase, which are generated from a one-step targeted attack and derived from a closed-form heuristic solution, respectively. Also considering perturbing components besides the images, **Adversarial Weight Perturbation** (AWP) [102] regularizes the flatness of weight loss landscape and robustness gap, using a double-perturbation mechanism that disturbs both inputs and weights. This defense boosts the robustness of multiple existing adversarial training methods, confirming that it can be applied to other methods. Since disturbing weights shows promising results, **Geometry-aware Instance-reweighted Adversarial Training** (GI-AT) [103] proposes using standard adversarial training considering that the weights are based on how difficult it is to attack a natural data point, *i.e.*, natural data points closer to class boundary are less robust.

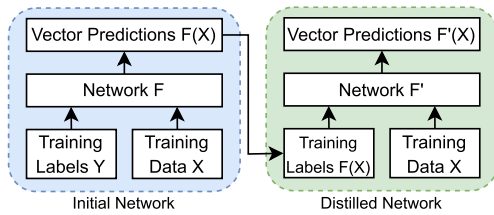


FIGURE 6. Method proposed by Defensive Distillation [104]. An Initial Network is trained on the dataset images and labels (discrete values). Then, the predictions given by the Initial Network are fed into another network, replacing the dataset labels. These predictions are continuous values, making the Distilled Network more resilient to adversarial attacks.

Summary: Adversarial Training is the most adopted approach to defend against adversarial attacks, demonstrating effective performance against specific attacks because they increase the boundaries for each class to include perturbed images. This type of defense consists in including adversarial examples in the training batches, using the original, other models, or heuristic labels, to train the robust model to recognize those images. Some works also suggest disturbing/changing the weights to be aware of the existence of adversarial examples. However, Adversarial Training using stronger attacks, such as PGD or C&W, is computationally expensive and lead to the search for more efficient approaches.

B. MODIFY THE TRAINING PROCESS

Defensive Distillation (DD) [104] uses knowledge distillation to train a resilient network by replacing the original labels with the labels of a previously trained network, as displayed in Figure 6, which is trained with original data and labels. To improve the results, Papernot and McDaniel [105] change the label vector used to train the distilled network by combining the original label with the first model uncertainty. Following the same context, **Robust Soft Label Adversarial Distillation (RSLAD)** [106] uses Robust Soft Labels (RSLs) produced by a teacher network to supervise the student training on natural and adversarial examples, without the latter having access to the original complex labels. **Low Temperature Distillation (LTD)** [107] considers low temperature in the teacher model using distillation to generate labels, by employing different fix temperatures for teacher and student models, which generates soft labels that can be integrated into existing works.

Robust Self-Training (RST) [108] uses a standard supervised approach to obtain pseudo-labels and feed them into a another network that targets adversarial robustness, bridging the gap between standard and robust accuracy using the unlabeled data. Exploring the results shown by not solely relying on original labels, **Self-Adaptive Training (SAT)** [109] uses as training labels a blend between the original labels and the model predictions, ignoring the first epochs due to instability, and updates their training labels throughout the training phase. In terms of changing the original data, **Visual Causal Feature Learning** [110]

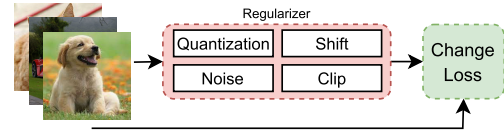


FIGURE 7. Schematic overview of the bit plane feature consistency [115]. This method applies multiple operations to input images, simulating adversarial images. Then, the loss is changed to include a regularizer (new term), which compares the original images with these manipulated images.

trains one network to predict causal effects and applies a manipulator function that return an image similar to the original one with the desired causal effect, performing data augmentation, which are fed to the classifier. **SENSEI** [111] and **SENSEI-SA** [111] use software testing methodologies to perform data augmentation, naively replacing each data point with a suitable variant or leaving it unchanged and selecting the opportunities suitable for skipping this augmentation process, respectively. Cutout [112] and Mixup [113] data augmentation techniques fail to prevent overfitting and make the model over-regularized, thus **Overfit Reduction (OR)** [114] uses a combination of early stopping and semi-supervised data augmentation to achieve substantial improvements.

Another considered approach is to add regularization terms, **Triplet Loss Adversarial (TLA)** [116] is a standard loss with an additional term that clusters natural and adversarial images of the same class and pushes the remaining classes further apart, using the Random Negative and the Switch Anchor techniques. **Curvature Regularization (CURE)** [117] introduces a regularizer term that penalizes large eigenvalues and minimizes the curvature of the loss surface, which is essential to achieve robustness. **Bit Plane Feature Consistency (BPFC)** [115] is a regularizer that uses the original and preprocessed images as shown in Figure 7, making the network give more importance to the higher bit planes. **Manifold Regularization for Locally Stable (MRLS)** [118] networks approximates the Laplacian graph when the data is sparse and exploits the continuous piece-wise linear nature of ReLU to create smooth decision boundaries.

Reducing the sensitive of neurons and their robust error are also valid approaches, **Controlling Neural Level Sets (CNLS)** [119] uses the level set values (e.g., decision boundaries) from a sample network to relate the sample position with the network parameters, modifying the decision boundaries of the network. **Sensitive Neuron Stabilizing (SNS)** [120] includes a regularizer dedicated to maximizing the similarities of sensitive neuron behaviors when evaluation clean and adversarial examples, stabilizing the sensitive neurons that significantly contribute to predictions under adversarial settings. **Self-Consistent Robust Error (SCORE)** [121] employs local equivariance to minimize the robustness error of a network, easing the reconciliation between robustness and accuracy and dealing with worst-case uncertainty. **Parametric Shifted Sigmoidal Linear Unit (PSSiLU)** [122] changes the activation function to create

high finite curvature and output positive results with negative inputs, creating a parameterized activation function, that is used with adversarial training.

Learning with a Strong Adversary (LSA) [123] uses adversarial learning, with the adversary applying perturbations to maximize the classification error and the classifier minimizing the misclassification error against the adversary, to significantly increase robustness while maintaining clean high accuracy. **Learnable Boundary Guided Adversarial Training (LBGAT)** [124] assumes that a model trained on clean data embeds the most discriminative features and constrains the logits from the robust model to make them similar to the first one. Recently, literature [125], [126], [127] demonstrated that neural Ordinary Differential Equations (ODE) are naturally more robust to adversarial attacks than vanilla networks. **Stable neural ODE for deFending against adversarial attacks (SODEF)** [128] uses optimization formulation to force the extracted feature points to be within the vicinity of Lyapunov-stable equilibrium points, suppressing the input perturbations.

Summary: This Modify the Training Process defense represents a wide variety of approaches that demonstrate increased robustness when applied to standard networks, which is divided into performing model distillation, augmenting data, changing the original labels, including a regularizer, constraining the weights, stabilizing neuron sensitive. The underlying idea is to modify the training process of the model, which requires that the training process is repeated to create a more robust model, translating into an additional effort to include this defenses in currently operating models.

C. USE OF SUPPLEMENTARY NETWORKS

The use of detector networks to avoid the transmission of adversarial examples to the classifier was one of the first defense mechanisms. **MagNet** [129] uses multiple detectors, trained on the reconstruction error, that are capable of detecting significantly and slightly perturbed examples based on probability divergence. In the same context, **Adversary Detection Network (ADN)** [130] is a network trained using adversarial images for a classifier that detects if the input example is adversarial or clean. Figure 8 displays a schematic overview of detector networks. Instead of training a network on adversarial examples, **Feature Squeezing (FSqueeze)** [131] uses multiple squeezing techniques, which reduce the search space, before giving the image to the classifier and compares the predictions of standard and squeezed classifiers to detect an adversarial example. **Detector Graph (DG)** [132] constructs a Latent Neighborhood Graph for each original image and uses Graph Neural Networks (GNNs) [133] to distinguish between original and adversarial examples, using embeddings of image pixel values.

Removing the adversarial noise from the images is another approach introduced by **High-level representation Guided Denoiser (HGD)** [17] trains a Denoising AutoEncoder [134]

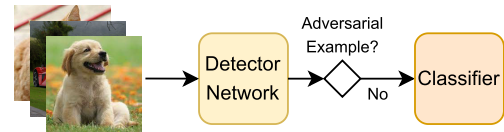


FIGURE 8. Schematic overview of the use of supplementary networks. The detector network was previously trained to detect adversarial images and is included between the input images and the classifier. This network receives the input images and determines if these images are adversarial or not. If they are not, they are redirected to the classifier; if they are, they are susceptible to human evaluation.

using the difference between original and adversarial images as the loss function and suppress the impact of adversarial examples. **Defense-GAN** [18] uses Wasserstein GANs (WGANs) [135] to accurately represent the set of clean training examples, which is attack-agnostic, and denoise the adversarial examples by minimizing the l_2 difference between the generator representation and the input image. **Embedding Regularized Classifier (ER-Classifier)** [136] uses an encoder and a discriminator to generate code vectors, reducing the dimensional space, and to distinguish these vectors from the ideal code vectors (sampled from a prior distribution), respectively, pushing the adversarial image into the clean image space.

Class Activation Feature-based Denoiser (CAFD) [137] is trained on samples generated by Class Activation Feature-based Attack to remove the noise from adversarial examples, by minimizing the distance between the class activation features and adversarial/natural examples. **Local Implicit Image Function (LIIF)** [138] trains an encoder to learn the construction of high resolution images, based on a coordinate and deep features around it, predicting the corresponding RGB value, and reduces the adversarial perturbation. **ADversarial defenSe with local impliCit functiOns (DISCO)** [139] is an encoder that predicts the clean RGB value for a pixel, which receives an adversarial image and a query pixel location, by employing a local implicit module function. **Reverse Attacks** [140] uses a self-supervised approach to find the suitable additive perturbations that repair the adversarial example and reduce the perturbations during the testing phase.

Summary: The use of Supplementary Networks assume that additional networks/transforms are included between the input and classifier model to either detect or reduce the adversarial noise. The benefit from this type of approaches is not requiring any change to the original classifier nor retraining, which can be applied to in production models. These additional networks usually learn the representation space of the original images to remove adversarial noise or detect if an image differs from this space, making them suitable to deal with unknown adversarial attacks.

D. CHANGE NETWORK ARCHITECTURE

To create inherently robust networks, Guo et al. [141] proposes the use of Neural Architecture Search to select which networks are more robust to adversarial attacks, named as **RobNets**, discovering that densely connected

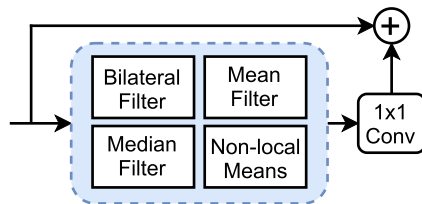


FIGURE 9. Overview of a feature denoising block [144], which can be included in the intermediate layers to make networks more robust. This method is an example of change network architecture.

patterns and convolution operations to direct connection edge are more robust. **HYDRA** [142] uses pruning techniques that are aware of the robust training objective, removing connections that negatively contribute to robustness, which creates compressed models with competitive standard and robust accuracy. **Dynamic Network Rewiring** (DNR) [143] creates pruned networks that have high robust and standard accuracy by unified constrained optimization formulation, using a hybrid loss function that merges model compression with robust adversarial training.

Zantedeschi et al. [145] discovered that Rectified Linear Units (ReLU) [146], [147] makes a small perturbation in the input accumulate with multiple layers and suggests the use of bounded ReLU (BReLU) [148] to prevent this accumulation. Benz et al. [149] found that BN [150] and other normalization techniques make CNNs more vulnerable to adversarial examples, suggesting that **Robust Features** are learned first and Non-Robust Features (which are the ones learned when using BN) afterward.

Feature Denoising (FD) [144] is a block that can be added to the intermediate layers of CNNs, consisting of feature-denoising operations, as described in Figure 9. **Input Random** [151] consists on adding layers at the beginning of CNNs that perform random resize, scaling the image to a random width and height, and random pad, padding with zeros around the resized image in a random manner. **Sparse Transformation Layer** (STL) [152] transforms the received images into a low-dimensional quasi-natural image space and can be included at the beginning of CNNs, reducing adversarial perturbations.

Summary: Different layers have distinct resilience to adversarial perturbations and some currently used techniques are more vulnerable. Multiple works propose the inclusion of specific layers to develop robust networks, yet require changes to the original architecture, hindering its application in production models.

E. PERFORM NETWORK VALIDATION

Datasets store their images using the Joint Photographic Experts Group (JPEG) [153] compression, Dziugaite et al. [154] evaluates the effect of JPEG, named as **JPG**, on the performance of FGSM, discovering JPEG compression can reverse smaller perturbations and does not have effect on perturbations with increased magnitude. Pang et al. [155] is the first to provide insights on

implementation details influence the model robustness, herein named as **Bag of Tricks** (BT), concluding that: 1) the robustness is significantly affected by weight decay; 2) early stopping of the adversarial attacks may deteriorate worst-case robustness; and 3) smooth activation benefits lower capacity models. **Fixing Data Augmentation** (FDA) [156] focuses on reducing the robust overfitting by using different data augmentation techniques, demonstrating that model weight averaging combined with data augmentation schemes can significantly increase robustness, which is enhanced when using spatial composition techniques. Gowal et al. [157] systematically studied the effect of multiple training losses, model sizes, activation functions, the addition of unlabeled data, and other aspects, concluding that larger models with Swish/SiLU [158] activation functions and model weight averaging can reliably achieve state-of-the-art results in robust accuracy.

DeepXplore [159] uses multiple networks as cross-referencing oracles (to avoid manual checking) and evaluates inputs that trigger different behaviors and achieve high neuron coverage as a joint optimization problem solved by gradient-based search techniques, which is the first automated white-box testing framework. In the context of automated testing, **DeepGauge** [160] identifies a testbed containing multi-faceted representations using a set of multi-granularity testing criteria and evaluates the resilience of networks using two different strategies, primary function and corner-case behaviors, considering neuron- and layer-level coverage criteria. **Surprise Adequacy for Deep Learning Systems** (SADL) [161] introduces the Surprise of an input, consisting of the network behavior difference between an input and the learned data, which is used in the adequacy criterion (Surprise Adequacy), a metric to ensure the input surprise range coverage.

In the context of formal verification, a tool [162], using Satisfiability Modulo Theory (SMT), for automatic **Safety Verification** of the classification assumes that a decision is safe when, after applying transformations in the input, the model decision does not change. Similarly, multiple works have been inspired on the SMT with some variations, such as exploiting the piece-wise linear structure [163], [164] and extending the interval of the theory [165]. However, the previous works lack scalability for larger networks, leading to the proposal of [166], [167], and [168] that use LaGrangian relaxation, l_∞ norm-bounded ball, and includes other components besides ReLU. **AI²** [169] and Mirman et al. [170] considered the abstract interpretation theory to represent the neural networks layers and activation functions, **PATROL** [171] uses game theory, and **Feature Partition Aggregation** [172] considers an ensemble of sub-models trained on different subsets.

Summary Software that is provided to the end consumer is often subject to a set of tests that validate its use in the real context, and it is a very important step to create a sense of security when using different devices. Similarly, the networks

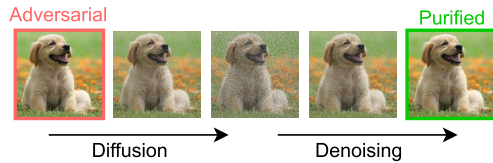


FIGURE 10. Overview of adversarial purification using denoising diffusion probabilistic models, adapted from [174]. The diffusion process is applied to an adversarial image, consisting of adding noise for a certain number of steps. During the denoising procedure, this noise is iteratively removed by the same amount of steps, resulting in a purified image (without perturbations).

should be subject to a set of tests that validate its performance in real context, using realistic attacks and images, ensuring their quality in real-world applications. These mechanisms provide a first insight on automated testing frameworks, formal verification, and implementation details that affect robustness, yet this topic remains highly unexplored.

F. ADVERSARIAL PURIFICATION

Adversarial Purification consists of defense mechanisms that remove adversarial perturbations using a generative model. **Improving Robustness Using Generated Data (IRUGD)** [173] explores how generative models trained on the original images can be leveraged to increase the size of the original datasets. Through extensive experiments, they concluded that Denoising Diffusion Probabilistic Models (DDPM) [174], a progression of diffusion probabilistic models [175], is the model that more closely resembles real data. Figure 10 presents the main idea behind the DDPM process.

Due to the great results in image synthesis displayed by the DDPM, Sehwag et al. [176] (**Proxy**) uses proxy distributions to significantly improve the performance of adversarial training by generating additional examples, demonstrating that the best generative models for proxy distribution are DDPM.

Inspired by previous works on adversarial purification [177], [178], **DiffPure** [179] uses DDPM for adversarial purification, receiving as input an adversarial example and recovering the clean image through a reverse generative process. Since this discovery, multiple improvements regarding the use of DDPM for Adversarial Purification have been studied. **Guided Diffusion Model for Adversarial Purification (GDMAP)** [180] receives as initial input pure Gaussian noise and gradually denoises it with guidance to an adversarial image.

DensePure [181] employs iterative denoising to an input image, with different random seeds, to get multiple reversed samples, which are given to the classifier and the final prediction is based on majority voting. Furthermore, Wang et al. [182] uses the most recent diffusion models [183] to demonstrate that diffusion models with higher efficiency and image quality directly translate into better robust accuracy.

Inspired by the need to evaluate the robustness of adversarial purification strategies, Lee et al. propose a gradual

noise-scheduling strategy that improves the robustness of diffusion-based purification.

Summary: Recently, Diffusion Models have been proposed as a new technique to generate images based on their training data, that iteratively adds random noise to the clean image and applies reverse transformations to remove the noise and generate a new image. This process allows to reconstruct a new image (purified), removing the added adversarial perturbations, which leads to great performance under different adversarial attacks and can be applied to in production models. This process is called Adversarial Purification and its the most promising defense mechanism when it can be applied in real-time.

VI. ADVERSARIAL EFFECTS ON VISION TRANSFORMERS

ViTs are susceptible to adversarial perturbations that alter a patch in an image [184], demonstrating higher robustness, almost double, compared with ResNet-50 [185].

To further evaluate the **robustness of ViT** to adversarial examples, Mahmood et al. [186] used multiple adversarial attacks designed for CNNs and found out that ViT has increased robustness (compared with ResNet) for the majority of the attacks and has no resilience to the C&W and MI-FGSM attacks. Additionally, to complement the results obtained from the performance of ViTs, an extensive study [187] using feature maps and attention maps explains this performance visually. Bai et al. [188] suggests that ViTs and CNNs are being unfairly evaluated and provides a fair and in-depth comparison between them, indicating that ViTs are as vulnerable to adversarial perturbations as CNNs.

Attacks. The transferability of adversarial examples from CNNs to ViTs was also evaluated, suggesting that examples from CNNs do not instantly transfer to ViTs, led to the proposal of **Self-Attention blended Gradient Attack (SAGA)** [186] to misclassify both ViTs and CNNs. The **Pay No Attention (PNA)** [189] attack, which ignores the gradients of attention, and the **PatchOut** [189] attack, which randomly samples subsets of patches, demonstrate high transferability. **Architecture-oriented Transferable Attacking (ATA)** [190] is a framework that generates transferable adversarial examples by considering the common characteristics among different ViT architectures, discovering that attentional patch-wise regions significantly influence the model decision. **Patch-fool** [191] proposes a dedicated attack framework that fools the self-attention mechanism by attacking a single patch with multiple attention-aware optimization techniques.

Change Architecture. Multiple works have adopted techniques used in CNNs architecture to include in ViTs, such as pyramid structure [192], “soft” convolutional inductive bias [193], and dimension reduction principle [194]. Additionally, Tokens-To-Token ViT [195] propose a new way to analyze the input by recursively aggregating neighboring Tokens into one Token, and Swin Transformer [196] introduce the representation of images using shifted windows.



FIGURE 11. Images from MNIST [208] and Fashion-MNIST [209] in the first five and last five columns, respectively.

Another issue with ViTs is the need to pretrain on large datasets and the time associated with it, which has been another research direction [197], [198]. **Robust Vision Transformer (RVT)** [199] reforms harmful components, proposing position-aware attention scaling and patch-wise augmentation, which can be applied to other ViTs. **Masked Autoencoder (MAE)** [200] partially masks the input tokens, with the visible patches being fed to the encoder and the masked patches added after the encoder and given to the decoder. Zhou et al. [201] discovered that self-attention promotes robustness through improved mid-level representations, proposing **Fully Attentional Networks (FANs)** which strengthen this capability. **Trade-off between Robustness and Accuracy of ViTs (TORA-ViTs)** [202] introduces two new components: a pair of accuracy and robustness adapters and a gated fusion module, which extract predictive and robust features and adjust their trade-off, respectively.

Adversarial Training. Smoothed ViTs [203] are trained using images composed of only one column of the original image, and the remaining are black, and the most predicted label is considered the correct one. **Pyramid Adversarial Training (PyramidAT)** [204] generates adversarial examples by perturbing the image at multiple scales and perform a “matched” Dropout and stochastic depth regularization, improving the robustness of ViTs. Gu et al. [205] evaluates the robustness of ViT to patch-wise perturbations, concluding that these models are more robust to naturally corrupted patches than CNNs while being more vulnerable to adversarially generated ones, proposing a simple **Temperature Scaling** based method that improves the robustness of ViTs. **Derandomized Smoothing** [206] uses a progressive smoothed image modeling task to train the ViTs, making them capture the more discriminating local context while preserving global semantic information, improving both robust and standard accuracy. **Token Gradient Regularization (TGR)** [207] reduces the variance of backpropagated gradient in internal blocks of ViTs in a token-wise manner and uses the regularized gradient to generate the adversarial examples.

Model Ensemble. PatchVeto [20] uses various transformers with different attention masks that output the encoding of the class and an image is only considered valid if all transformers reach a consensus in the voted class.

Summary: ViTs are a new technique, based on Transformers, to classify images that have similar or superior performance than CNNs and are more efficient. ViTs are also vulnerable to adversarial perturbations, yet there is not sufficient research on how susceptible are ViTs to the known adversarial perturbations and there are not many attacks designed specifically for ViTs. To the date it is known that the adversarial attacks designed for CNNs are not that

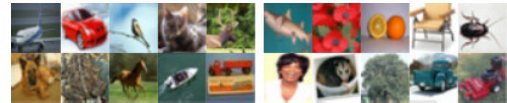


FIGURE 12. Images from CIFAR-10 [210] and CIFAR-100 [210] in the first five and last five columns, respectively.

easily transferred to ViTs, justifying further research on this topic, and changing the input and using techniques from CNN architecture display increased robustness. This section answers the proposed research question RQ1.

VII. DATASETS

One of the most used datasets is the **MNIST** [208] dataset, which contains images of handwritten digits collected from approximately 250 writers in shades of black and white, withdrawn from two different databases. Xiao et al. propose the creation of the **Fashion-MNIST** [209] dataset by using figures from a fashion website, which has a total size of 70,000 images, contains 10 classes, uses greyscale images, and each image has a size of 28×28 . The MNIST and Fashion-MNIST datasets are divided into train and test sets, containing 60,000 and 10,000 examples, respectively.

Fig. 11 displays the 10 digits (from 0 to 9) from the MNIST dataset in the first five columns and the 10 fashion objects from Fashion-MNIST dataset in the last five columns. MNIST is one of the most widely studied datasets in the earlier works of adversarial examples, with defense mechanisms already displaying high robustness and accuracy on this dataset. The same does not apply to Fashion-MNIST, which requires further analysis, despite having similar characteristics to MNIST.

Another widely studied dataset is the CIFAR-10, which, in conjunction with the CIFAR-100 dataset, are subsets from a vast database containing 80 million tiny images [211], 32×32 , and three color channels 75,062 different classes. **CIFAR-10** [210] contains only ten classes from this large database, with 6,000 images for each class, distributed into 50,000 training images and 10,000 test images. This dataset considers different objects, namely, animals and vehicles, usually found in different environments. **CIFAR-100** [210] contains 100 classes with only 600 images for each one with the same size and amount of color channels as the CIFAR-10 dataset. CIFAR-100 groups its 100 classes into 20 superclasses, located in different contexts/environments, making this dataset much harder to achieve high results.

Examples from the CIFAR-10 dataset are shown in Fig. 12 in the first five columns, and the remaining columns display examples of the superclasses from CIFAR-100. Due to the unsatisfactory results demonstrated by models trained on CIFAR-10, the CIFAR-100 dataset has not been included in most studies under the context of adversarial examples, suggesting that solving the issue of adversarial-perturbed images is still at its inception.

The **Street View House Numbers (SVHN)** [212] dataset provides the same challenge as MNIST: identifying which digits are present in a colored image, containing 10 classes



FIGURE 13. Images from street view house numbers [212] and german traffic sign recognition benchmark [213] in the first three and last three columns, respectively.

and an image size of 32×32 centered around a single character, with multiple digits in a single image. Regarding the dataset size, it has 630,420 digit images, but only 73,257 images are used for training, 26,032 images are used for testing, and the remaining 531,131 images can be used as additional training data. **German Traffic Sign Recognition Benchmark (GTSRB)** [213] is a dataset containing 43 classes of different traffic signs, demonstrating realistic scenarios, has 51,840 images, whose size varies from 15×15 to 222×193 , divided into training, validation, and test sets with 50%, 25%, and 25%, respectively, of the total images.

The difficulties associated with the SVHN dataset are displayed in the first three rows of Fig. 13, showing unique digits that occupy the whole image and multiple digits on different backgrounds. The same figure presents the different types of traffic signs in the GTSRB dataset, such as prohibition, warning, mandatory, and end of prohibition. These datasets require further analysis, namely the SVHN, which has the same task as MNIST yet preliminaries results show that its performance is inferior. GTSRB has realistic scenarios that might already contain of realistic adversarial examples.

ImageNet [214] is one of the largest datasets for object recognition, containing 1,461,406 colored images and 1,000 classes, with images being resized to 224×224 . This dataset collected photographs from Flickr, and other search engines, divided into 1.2 million training images, 50,000 validation images, and 100,000 test images. A possible alternative to ImageNet, when the dataset size is an important factor, is called **Tiny ImageNet** [215], a subset of ImageNet that contains fewer classes and images. This dataset contains only 200 classes (from the 1,000 classes in ImageNet), 100,000 training images, 10,000 validation images, and 10,000 test images. These classes include animals, vehicles, household items, insects, and clothing, considering the variety of contexts/environments that these objects can be found. Their images have a size of 64×64 and are colored.

ImageNet-A [216] is a subset of ImageNet, containing only 200 classes from the 1,000 classes, covering the broadest categories in ImageNet. ImageNet-A is a dataset composed of real-world adversarially filtered images, which were obtained by deleting the correctly predicted images by ResNet-50 classifiers, demonstrating transferability to unseen models, which makes this dataset suitable for evaluating the robustness of multiple classifiers. Two additional benchmarks, **ImageNet-C** [217] and **ImageNet-P** [217], were designed to evaluate

TABLE 3. Relevant characteristics to the context of adversarial examples of the mentioned datasets. #classes means the number of classes in the dataset. Empty color column means that the images in that dataset use grayscale or black and white shades. Datasets with * are only used for testing.

Dataset	Size	#Classes	Classes	Color
MNIST	70,000	10	Digits	
Fashion-MNIST	70,000	10	Clothing	
CIFAR-10	60,000	10	Animals Vehicles	✓
SVHN	630,420	10	Digits	✓
GTSRB	51,840	43	Traffic Signs	✓
CIFAR-100	60,000	100	Household Items Outdoor Scenes	✓
Tiny ImageNet	120,000	200	Animals Household Items	✓
ImageNet-A*	7,500	200	Vehicles Food	✓
ImageNet-C*	3,750,000	200	Vehicles Food	✓
ImageNet-P*	15,000,000	200	Vehicles Food	✓
ImageNet	1,431,167	1,000	Vehicles Electronic devices	✓
ImageNet-CD*	736,515	1,000	Vehicles Electronic devices	✓

the robustness of DNNs. The ImageNet-C standardizes and expands the corruption robustness topic, consisting of 75 corruptions applied to each image in the ImageNet validation set. ImageNet-P applies distortions to the images, though it differs from ImageNet-C because it contains perturbation sequences using only 10 common perturbations. Another benchmark is the **ImageNet-COLORDISTORT** (ImageNet-CD) [218], which considers multiple color distortions using different color space representations. This dataset contains the 1,000 classes from ImageNet, removing images without color channels, and the same image considers multiple color distortions under the Red Green Blue (RGB), Hue-Saturation-Value (HSV), CIELAB, and YCbCr color spaces considered common transformations used in image processing.

It is possible to observe a set of images withdrawn from ImageNet in the top left of Fig. 14. Additionally, some images misclassified by multiple classifiers (ImageNet-A) are shown in the top right of the same figure. The bottom represents the ImageNet with common corruptions and perturbations and is manipulated by multiple image techniques on the left and right, respectively. Table 3 summarizes the main characteristics of the datasets presented throughout this section. Recently, works are presenting their results in Tiny ImageNet dataset to estimate the performance of adversarial attacks on ImageNet dataset.

VIII. METRICS AND STATE-OF-THE-ART RESULTS

This section compiles the most important metrics that are applied in the context of adversarial attacks for Object Recognition and the state-of-the-art results for both CNNs and ViTs under multiple attack settings. Note that some of

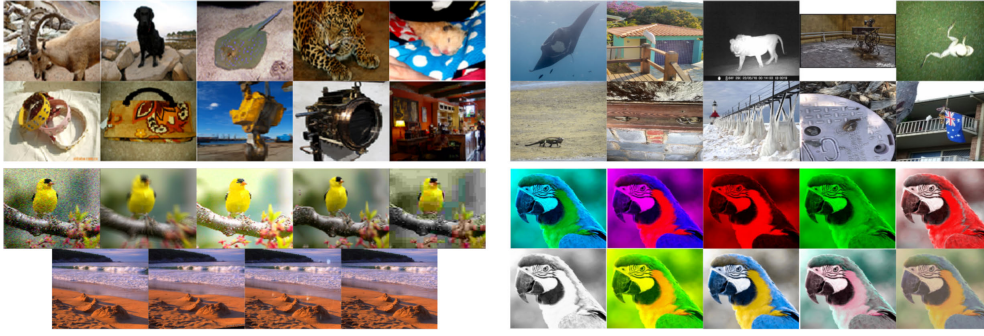


FIGURE 14. Images from ImageNet [214] in the top left, from ImageNet-A [216] in the top right, from ImageNet-C and ImageNet-P datasets [217] in the bottom left, and ImageNet-COLORDISTORT [218] in the bottom right.

these metrics were not specifically designed for adversarial attacks, but they are used throughout the literature.

A. EVALUATION METRICS

1) ACCURACY

This metric measures the number of samples that are correctly predicted by the model, which is defined as:

$$\text{accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}, \quad (7)$$

where TP refers to True Positive, TN to True Negative, FP to False Positive, and FN to False Negative. When considering original images, this metric is denominated as *Clean Accuracy* and, when using adversarial images, is named as *Robust Accuracy*. Accuracy allows the comparison with previously proposed methods and clearly displays the performance of the model under clean and attack scenarios.

2) FOOLING RATE

The expected number of times the attack is able to flip the label, known as fooling rate, denoted by FR [219], measures the percentage of success for an adversarial attack, formally defined [220]:

$$\text{FR} = \frac{\sum_{i=1}^N (y(i) \neq \hat{y}(i))}{N}, \quad (8)$$

where $y(i)$ and $\hat{y}(i)$ refer to the pre-attack label and post-attack label associated with image i and N is the number of images. The Fooling Rate is an attack-specific metric that can measure how effectively the attack is on different images.

3) AVERAGE ROBUSTNESS

To objectively evaluate the robustness to adversarial perturbations of a classifier f , the average robustness $\hat{p}_{\text{adv}}(f)$ is defined as [28]:

$$\hat{p}_{\text{adv}}(f) = \frac{1}{D} \sum_{x \in D} \frac{\|\hat{r}(x)\|_2}{\|x\|_2}, \quad (9)$$

where $\hat{r}(x)$ is the estimated minimal perturbation obtained using the attack, and D denotes the test set.

TABLE 4. Accuracy comparison of different defense mechanisms on CIFAR-10 under auto-attack attack, l_∞ and $\epsilon = 8/255$, partially adapted from [139]. Clean and robust refers to accuracy without and with adversarial attacks, respectively.

Architecture	Defense Method	Year	Accuracy	
			Clean	Robust
WRN28-10	Input Random [151]	2017	94.3	8.6
	BAT [101]	2019	92.8	29.4
	FS [19]	2019	90.0	36.6
	Jpeg [154]	2016	83.9	50.7
	Pretrain [221]	2019	87.1	54.9
	UAT [84]	2019	86.5	56.0
	MART [91]	2020	87.5	56.3
	HYDRA [142]	2020	89.0	57.1
	RST [108]	2019	89.7	59.5
	GI-AT [103]	2020	89.4	59.6
	Proxy [176]	2021	89.5	59.7
	AWP [102]	2020	88.3	60.0
	FDA [156]	2021	87.3	60.8
	HAT [85]	2021	88.2	61.0
	SCORE [121]	2022	88.6	61.0
	PSSiLU [122]	2022	87.0	61.6
	Gowal <i>et al.</i> [157]	2020	89.5	62.8
	IRUGD [173]	2021	87.5	63.4
	Wang <i>et al.</i> [182]	2023	92.4	67.3
	STL [152]	2019	82.2	67.9
	DISCO [139]	2022	89.3	85.6
WRN34-10	Free-AT [92]	2019	86.1	41.5
	AT-PGD [41]	2018	87.1	44.0
	YOPO [93]	2019	87.2	44.8
	TLA [116]	2019	86.2	47.4
	LAT [94]	2019	87.8	49.1
	SAT [97]	2020	86.8	50.7
	FAT [96]	2022	85.3	51.1
	LBGAT [124]	2021	88.2	52.3
	TRADES [88]	2019	84.9	53.1
	SAT [109]	2020	83.5	53.3
	Friend-AT [98]	2020	84.5	55.5
	AWP [102]	2020	85.4	56.2
	LTD [107]	2021	85.2	56.9
	OA-AT [99]	2021	85.3	58.0
	Proxy [176]	2022	86.7	60.3
	HAT [85]	2021	91.5	62.8
	SCORE [121]	2022	89.0	63.4
WRN-70-16	IRUGD [173]	2021	91.1	65.9
	Gowal <i>et al.</i> [157]	2020	88.7	66.1
	FDA [156]	2021	92.2	66.6
	Wang <i>et al.</i> [182]	2023	93.3	70.7
	SODEF [128]	2021	93.7	71.3

4) DESTRUCTION RATE

To evaluate the impact of arbitrary transformations on adversarial images, the notion of destruction rate d is introduced and formally defined as [58]:

$$d = \frac{\sum_{k=1}^n C(\mathbf{X}^k, y_{\text{true}}^k) - C(\mathbf{X}_{\text{adv}}^k, y_{\text{true}}^k) C(T(\mathbf{X}_{\text{adv}}^k), y_{\text{true}}^k)}{\sum_{k=1}^n C(\mathbf{X}^k, y_{\text{true}}^k) C(\mathbf{X}_{\text{adv}}^k, y_{\text{true}}^k)}, \quad (10)$$

TABLE 5. Accuracy comparison of different defense mechanisms on CIFAR-10 under PGD attack, l_∞ and $\epsilon = 8/255$, results drawn from each paper. Clean and Robust refers to accuracy without and with adversarial attacks, respectively. Defenses with “–” on clean accuracy do not have a clean accuracy reported.

Defense Method	Year	Architecture	Accuracy	
			Clean	Robust
BPFC [115]	2020	ResNet-18	82.4	34.4
SNS [120]	2021	VGG-16	86.0	39.6
AT-MIFGSM [82]	2017	Inception v3	85.3	45.9
AT-PGD [41]	2018	ResNet-18	87.3	47.0
RobNets [141]	2020	RobNet-free	82.8	52.6
HGD [17]	2018	DUNET	92.4	53.1
RSLAD [106]	2021	ResNet-18	83.4	54.2
MART [91]	2020	WRN-28-10	83.1	55.6
TRADES [88]	2019	WRN-34-10	84.9	56.4
BagT [155]	2020	WRN-34-10	–	56.4
RO [114]	2020	ResNet-18	–	56.8
DOA [87]	2019	VGGFace	93.6	61.0
AWP [102]	2020	WRN-28-10	–	63.6
FS [19]	2019	WRN-28-10	90.0	68.6
CAFD [137]	2021	DUNET	91.1	87.2

TABLE 6. Accuracy comparison of different defense mechanisms on CIFAR-100 under auto-attack attack, l_∞ and $\epsilon = 8/255$, partially adapted from [139]. Clean and robust refers to accuracy without and with adversarial attacks, respectively.

Architecture	Defense Method	Year	Accuracy	
			Clean	Robust
WRN28-10	Input Random [151]	2017	73.6	3.3
	LIIF [138]	2021	80.3	3.4
	Bit Reduction [131]	2017	76.9	3.8
	Pretrain [221]	2019	59.2	28.4
	SCORE [121]	2022	63.7	31.1
	FDA [156]	2021	62.4	32.1
	Wang <i>et al.</i> [182]	2023	78.6	38.8
	Jpeg [154]	2016	61.9	39.6
	STL [152]	2019	67.4	46.1
	DISCO [139]	2022	72.1	67.9
WRN34-10	SAT [97]	2020	62.8	24.6
	AWP [102]	2020	60.4	28.9
	LBGAT [124]	2021	60.6	29.3
	OA-AT [99]	2021	65.7	30.4
	LTD [107]	2021	64.1	30.6
	Proxy [176]	2022	65.9	31.2
WRN-70-16	DISCO [139]	2022	71.6	69.0
	SCORE [121]	2022	65.6	33.1
	FDA [156]	2021	63.6	34.6
	Gowal <i>et al.</i> [157]	2020	69.2	36.9
	Wang <i>et al.</i> [182]	2023	75.2	42.7

where n is the number of images, $\mathbf{X}^k$ is the original image from the dataset, y_{true}^k is the true class of this image, $\mathbf{X}_{\text{adv}}^k$ is the adversarial image corresponding to that image, and T is an arbitrary image transformation. $\neg C(\mathbf{X}_{\text{adv}}^k, y_{\text{true}}^k)$ is defined as the binary negation of $C(\mathbf{X}_{\text{adv}}^k, y_{\text{true}}^k)$. Finally, the function $C(\mathbf{X}, y)$ is defined as [58]:

$$C(\mathbf{X}, y) = \begin{cases} 1, & \text{if image } \mathbf{X} \text{ is classified as } y; \\ 0, & \text{otherwise.} \end{cases} \quad (11)$$

This metric can evaluate how resilient the perturbations are even when susceptible to multiple common transformations. This subsection answers the proposed research question RQ2.

B. DEFENSE MECHANISMS ROBUSTNESS

The most commonly used metric to evaluate models is accuracy, which evaluates the results on both original (**Clean Accuracy**) and adversarially perturbed (**Robust Accuracy**) datasets. One of the earliest and strongest adversarial attacks

proposed was PGD, which is used by multiple defenses to evaluate their robustness. Table 5 displays defenses evaluated on CIFAR-10 under multiple steps PGD attack, ordered by increasing robustness. For the PGD attack, the best performing defenses are from approaches that use supplementary networks (CAFD) or modify the training process (FS and AWP). Overall, Wide ResNets [222] have better robust accuracy, due to high-capacity networks exhibiting greater adversarial robustness [41], [82], suggesting the usage of these networks in future developments of adversarial attacks and defenses.

To assess the robustness of defenses for white and black-box settings, Auto-Attack has gained increased interest over PGD in recent works. Tables 4, 6, and 7 present a set of defenses that are evaluated under Auto-Attack, on CIFAR-10, CIFAR-100, and ImageNet, respectively, ordered by increasing Robust Accuracy. The most used networks are Wide ResNets with different sizes, with the biggest Wide ResNet displaying better results overall, and the most resilient defense derives from the use of supplementary networks (DISCO), followed by modifying the train process (SODEF) and changing network architecture (STL). The results suggest that the inclusion of additional components to sanitize inputs of the targeted model (use of supplementary networks) is the most resilient approach for model robustness in white and black-box settings. The updated results for defenses under Auto-Attack can be found on the RobustBench [223] website.

We also provide results for ViTs on ImageNet-A dataset in Table 8, considering Vanilla ViTs and Robust ViTs, ordered by increasing Robust Accuracy. It is possible to observe that there is still a great margin of improvement for robustness in ViTs, achieving the best results when using a trade-off between accuracy and robustness (TORA-ViT-B/16) and partially occluding the input image (MAE-ViT-B). Finally, to provide a effortless introduction and exploration into the adversarial attacks topic, we collect the most used libraries in Python, implementations of the discussed attacks and defenses, presenting the respective GitHub URL, in Tables 9, 10, and 11. This subsection answers the proposed research question RQ3.

IX. PRACTICAL APPLICATIONS AND FUTURE DIRECTIONS

A. PRACTICAL APPLICATIONS

1) AUTONOMOUS DRIVING

Zhang *et al.* [225] proposes an end-to-end driving safety evaluation framework with designed performance metrics for the assessment of driving safety, suggesting that it is desirable to evaluate the robustness of deep learning models in terms of driving safety instead of model precision. Rossolini *et al.* [226] focuses on evaluating the robustness of segmentation models to real-world adversarial attacks and demonstrates that these already have a threshold of robustness. Wang *et al.* [227] explored how the manipulation of 3D point clouds can affect autonomous driving and a deeper analysis regarding adversarial reinforcement learning

TABLE 7. Accuracy comparison of different defense mechanisms on ImageNet under auto-attack attack, l_∞ and $\epsilon = 4/255$, partially adapted from [139]. Clean and robust refers to accuracy without and with adversarial attacks, respectively.

Architecture	Defense Method	Year	Accuracy	
			Clean	Robust
ResNet-18	Bit Reduction [131]	2017	67.6	4.0
	Jpeg [154]	2016	67.2	13.1
	Input Random [151]	2017	64.0	17.8
	Salman <i>et al.</i> [224]	2020	52.9	25.3
	STL [152]	2019	65.6	32.9
	DISCO [139]	2022	68.0	60.9
ResNet-50	Bit Reduction [131]	2017	73.8	1.9
	Input Random [151]	2017	74.0	18.8
	Cheap-AT [95]	2020	55.6	26.2
	Jpeg [154]	2016	73.6	33.4
	Salman <i>et al.</i> [224]	2020	64.0	35.0
	STL [152]	2019	68.3	50.2
WRN-50-2	DISCO [139]	2022	72.6	68.2
	Bit Reduction [131]	2017	75.1	5.0
	Input Random [151]	2017	71.7	23.6
	Jpeg [154]	2016	75.4	24.9
	Salman <i>et al.</i> [224]	2020	68.5	38.1
	DISCO [139]	2022	75.1	69.5

is required. For further details on attacks and defenses in the context of autonomous driving, we refer to a survey on this specific topic by Deng et al. [228].

2) MEDICAL IMAGING

Bortsova et al. [229] performed extensive experiments that show pretraining on ImageNet makes the models more vulnerable to transferability and the disparity between the data and architecture of the target and the substitute model can reduce the success of attacks. Tsai et al. [230] shows that all medical imaging is vulnerable to one-pixel attack and its effect would have devastating effects on the diagnosis of diseases. Kaviani et al. [231] and Apostolidis et al. [232] survey existing adversarial attacks and defenses in the context of medical imaging, emphasizing the urgent need to develop resistant mechanisms.

3) AUTHENTICATION

Currently, different methods are used to authenticate an user, giving it access to a device or location. However, these mechanisms also use artificial intelligence to evaluate if the person being authenticated has permissions or not. Tan et al. [233] evaluate how adversarially generated behavioral mouse dynamics can be used to bypass authentication mechanisms. Fingerprints are other example of authentication susceptible to adversarially crafted examples, using different methods [234], [235], [236]. Another daily-basis authentication mechanism is face recognition, which is vulnerable to more realistic attacks [237]. Wang et al. [238] provides a classification for attacks in authentication, reviewing common adversarial attacks and defenses.

4) OTHER AREAS

Natural Language Processing, which uses textual data, is also susceptible to adversarial attacks [239], with specific focus on *Social Network Applications* [240] and on Large Language Models, which have high relevance due to their broad scope.

TABLE 8. Accuracy comparison of different models and defense mechanisms on ImageNet-A, partially adapted from [202]. Clean and robust refers to accuracy without and with adversarial attacks, respectively.

Category	Model/Defense Method	Year	Accuracy	
			Clean	Robust
Vanilla ViT	ViT-B/16 [27]	2020	72.8	8.0
	ViT-B/16+MixUp [27]	2020	77.8	12.2
	ViT-B/16+CutMix [27]	2020	75.5	14.8
	ViT-B/16+AugReg [198]	2021	79.9	17.5
	PVT-Large [192]	2021	81.7	26.6
	DeiT-B/16 [197]	2021	82.0	27.4
	T2T-ViT _{r-24} [195]	2021	82.6	28.9
	ConViT-B [193]	2021	82.4	29.0
	PiT-B [194]	2021	82.4	33.9
	Swin-B [196]	2021	83.4	35.8
Robust ViT	PyramidAT [204]	2022	81.7	23.0
	RVT-B [199]	2022	82.7	28.5
	FAN-L-ViT [201]	2022	83.9	34.2
	MAE-ViT-B [200]	2022	83.6	35.9
	TORA-ViT-B/16 [202]	2023	84.1	46.5

Financial Transactions are also a critical area that employs artificial intelligence, which is susceptible to adversarial attacks [241], [242]. *Smart Grids* that intelligently manage the distribution of electricity throughout a city are also vulnerable to adversarial attacks [243].

B. FUTURE DIRECTIONS

Following the *de facto* standards adopted by the literature, we suggest that future proposals of **defense mechanisms should be evaluated on Auto-Attack**, using the **robust accuracy** as a metric, and **Wide ResNets as standard architecture**. The adversarial defense that demonstrates better results is **Adversarial Training**, which should be a **requirement when evaluating attacks and defenses**. Additionally, we believe that standardization will directly degenerate into the increased efficacy of researching in the context of adversarial attacks.

The state-of-the-art results show that MNIST and CIFAR-10 datasets are already saturated. Other datasets should be further evaluated, namely: 1) **CIFAR-100 and ImageNet** since adversarial defenses do not achieve state-of-the-art clean accuracy (91% and 95%, respectively); 2) **GTSRB and SVHN**, depicting harder scenarios with greater variations of background, inclination, and luminosity; and 3) **Fashion-MNIST** that would allow better comprehension of which image properties influence DNNs performance. There should be an effort to **develop a metric/process to evaluate the amount of distortion** caused by new adversarial attacks, which would ease the expansion of adversarial attacks to other datasets that do not have a standardized ϵ value.

White-box attacks consider that the adversary has access to the network and training data, yet this is not feasible in real contexts, translating into the need of **focusing more on the development of black-box attacks**. A unique black-box set, **physical attacks, also require additional evaluation**, considering the properties of the real world and perturbations commonly found in it. Considering the increasing liberation of ML in the real world, end-users can partially control the training phase of DNNs, suggesting that **gray-box attacks**

TABLE 9. Most commonly used python libraries, containing multiple adversarial attacks and defenses, ordered by date of last update.

Name	Description	PT	TF	K	MX	Scikit	JAX	Last Update	Code
Torchattacks	Adversarial Attacks	✓						October 2023	URL
RobustBench	Datasets, Models, and Adversarial Attacks	✓						September 2023	URL
Adversarial Robustness Toolbox	Metrics, Models, and Adversarial Attacks		✓	✓		✓		September 2023	URL
Advbox	Adversarial Attacks and Defenses	✓	✓	✓	✓			August 2022	URL
Foolbox	Adversarial Attacks	✓	✓				✓	May 2022	URL
advertorch	Adversarial Attacks and Defenses	✓						May 2022	URL

TABLE 10. Official and unofficial (identified with *) source code of discussed adversarial attacks.

Type	Name	PT	TF	Code
White-box Attacks	FGSM* [21]	✓		URL
	DeepFool [28]	✓		URL
	SF [29]	✓		URL
	SIAA [40]	✓	✓	URL
	PGD [41]		✓	URL
	MI-FGSM [42]		✓	URL
	C&W [43]		✓	URL
	MGAA [44]		✓	URL
	AdvGAN* [45]	✓		URL
	BIM* [58]	✓		URL
Universal Adversarial Perturbations	UAP [53]		✓	URL
	UAN [54]	✓		URL
	SGA [55]	✓		URL
	TRM-UAP [56]	✓		URL
	AdvEncoder [57]	✓		URL
	AdvCam [59]		✓	URL
Black-box Attacks	Auto-Attack [25]	✓		URL
	LQIA [62]		✓	URL
	GRA [68]		✓	URL
	SIA [69]	✓		URL
	Square Attack [70]	✓		URL
	StyLess [72]	✓		URL
	FIA [73]		✓	URL
	MSA [75]	✓		URL
	AdaEA [76]	✓		URL

will intensify (access only to network or data). Recent developments [47] point out to **UAPs that break the l_p bound but are semantically realistic** should be further analyzed.

The different network architectures are designed to increase the clean accuracy of DNNs in particular object recognition datasets, yet there should be **further evaluation on the impact of the different layers and their structure**. Additionally, there should be a focus on ensuring robustness through the **application of formal verification techniques**. ViTs introduce a new paradigm in image analysis and are more robust against natural corruptions, suggesting that **building ViT inherently robust to adversarial examples** might be a possible solution.

One of the widely adopted approach is the usage of supplementary networks that detect the existence of adversarial attacks, yet a **greater focus on evaluating the robustness of detector networks** is another issue that hinders the deployment of these defenses. DDPM are generative models that perform adversarial purification of images, but they can not be applied in real-time since they take up to dozens of seconds to create a single purified image. Therefore, an effort on **developing close to real-time adversarial purification strategies** is a viable strategy for future works.

The common factor among all the scenarios in adversarial attacks is the existence of an attacker and the

TABLE 11. Official and unofficial (identified with *) source code of discussed adversarial defenses.

Type	Name	PT	TF	K	MX	Code
Adversarial Training	FScatter [19]	✓				URL
	RobDNN* [80]	✓				URL
	AdvScale* [82]		✓			URL
	EAT* [83]		✓			URL
	UAT [84]		✓			URL
	HAT [85]	✓				URL
	DOA [87]	✓				URL
	TRADES [88]	✓				URL
	MART [91]	✓				URL
	Free-AT [92]	✓	✓			URL
	YOPO [93]	✓				URL
	LAT [94]		✓			URL
	Cheap-AT [95]	✓				URL
	Friend-AT [98]	✓				URL
	AWP [102]	✓				URL
	GI-AT [103]	✓				URL
Modify the Training Process	DD* [104]	✓				URL
	E-DD* [105]	✓				URL
	RSLAD [106]	✓				URL
	RST [108]	✓				URL
	SAT [109]	✓				URL
	SENSEI [111]		✓			URL
	OR [114]	✓				URL
	TLA [116]		✓			URL
	CR* [117]	✓				URL
	BPFC [115]	✓				URL
	MRLS [118]	✓				URL
	CNLS [119]	✓				URL
	SCORE [121]	✓				URL
	LSA [123]				✓	URL
	LBGAT [124]	✓				URL
	SODEF [128]	✓				URL
Use of Supplementary Networks	HGD [17]	✓				URL
	Defense-GAN [18]			✓		URL
	MagNet* [129]			✓		URL
	FSqueez [131]		✓	✓		URL
	LIIF [138]	✓				URL
	DISCO [139]	✓				URL
Change Network Architecture	ReverseA [140]	✓				URL
	RobNets [141]	✓				URL
	HYDRA [142]	✓				URL
	DNR [143]	✓				URL
	FD [144]		✓			URL
	EffAdv* [145]			✓		URL
Perform Network Validation	Benz <i>et al.</i> [149]	✓				URL
	InputRand [151]		✓			URL
	STL [152]			✓		URL
	BT [155]	✓				URL
	FDA [156]	✓				URL
	Gowal <i>et al.</i> [157]	✓				URL
Adversarial Purification	DeepXplore [159]		✓			URL
	DeepGauge [160]		✓			URL
	SADL [161]			✓		URL
	SafeVeri [162]			✓		URL
	IRUGD [173]	✓				URL
	Proxy [176]	✓				URL
Adversarial Purification	SOAP [177]	✓				URL
	Yoon <i>et al.</i> [178]	✓				URL
	DiffPure [179]	✓				URL
	DensePure [181]	✓				URL
	Wang <i>et al.</i> [182]	✓				URL
	EDM [183]	✓				URL

removal/withdrawal of attackers from the network [244] would reduce the chance of security breaches. This section answers the proposed research question RQ4.

X. CONCLUSION

Networks are vulnerable to a set of inputs, denominated as adversarial examples, that modify their output and are constructed by adding a perturbation to the original image. This survey presents background concepts, such as adversary capacity and vector norms, and provides a comparison with existing surveys in the literature. Adversarial attacks are organized based on the adversary knowledge, highlighting the emphasis toward white box settings, and adversarial defenses are clustered into six domains, with adversarial training being the most popular strategy. We also present the latest developments of adversarial settings in ViTs, describe the commonly used datasets and metrics, providing results in CIFAR-10, CIFAR-100, and ImageNet, and provide the source code for adversarial attacks and defenses, finishing with a set of open issues that remain less explored.

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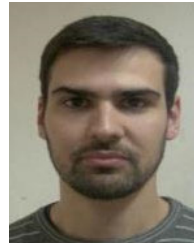
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